

PRODUCT**KERATO PSOR SHAMPOO**

PRIMARY PACKAGING	200 ml BOTTLE
CATEGORY	DERMOCOSMETIC
ACTIVE INGREDIENTS	KERATOLYTIC ACTION UREA 10% Exfoliating and keratolytic action. Present in the epidermis, it essentially comes from the sweat glands and is part of the hygroscopic compounds of the stratum corneum, which constitute the natural hydration factor (Natural Moisturizing Factor). Urea facilitates the elimination of scales, favors the renewal of corneocytes and constitutes a first choice moisturizing and humectant agent for the epidermis since it is able to break the hydrogen bonds of the protein chains and to alternate their configuration and aggregation. Consequently, at high concentrations (above 10%), it has a marked keratolytic effect and increases skin permeability. MALIC ACID 2% Alpha hydroxy acid of natural origin with exfoliating, lightening and antioxidant properties. Alpha and beta hydroxyacids are often suitably associated and used in different concentrations, to enhance their ability to delicately and progressively remove the outermost layers of the epidermis, promoting the removal of flaky corneocytes, i.e. the outermost, old and dead skin cells. dehydrated, which give the skin a dull and lifeless appearance. Alpha Hydroxy Acids in particular, promote cell turnover and skin renewal through a superficial exfoliation of the skin and a stimulus on the synthesis of glycosaminoglycans (or GAGs, or mucopolysaccharides), with a consequent increase in the production of new collagen and new elastin. SALICYLIC ACID 2% Beta hydroxy acid. The molecule derives from the enzymatic hydrolysis of salicin, a salicoside glucoside extracted from willow (<i>Salix alba</i>) and is found in nature in numerous plants in the form of esters. Salicylic acid is used in the cosmetic and dermatological fields for its keratolytic and smoothing properties. When applied to the skin, in fact, it performs a powerful keratolytic action, as it is able to selectively break the protein chain of keratin, abundantly present in the corneumocytes, thus decreasing the cohesive forces that unite these cells to each other. In this way, the desquamation of the keratinocytes is encouraged while at the same time encouraging the cell renewal. In addition to the keratolytic and smoothing properties, salicylic acid also has interesting properties bacteriostatic and soothing properties: the substance, in fact, rapidly penetrates the inflamed lesions by exercising excellently these functions, without however causing evident irritation phenomena.

ANTI-INFLAMMATORY AND ANTIOXIDANT ACTION

NICOTINAMIDE (VITAMIN PP) 2%

Water-soluble vitamin well known for its multiple applications, particularly in the dermatological field. It has a marked anti-inflammatory power and it is assumed that the mechanism of action is to be found in its antihistamine activity, scavenger of free radicals and inhibition of the enzyme 3.5 cyclic AMP-phosphodiesterase.

GLYCYRRETHINIC ACID 0.2%

The anti-inflammatory activity (Marjan Nassiri Asl et al., 2008) basically depends on two mechanisms:

- 1) glycyrrhetic acid has a planar structure very similar to the cortisone molecule, it can therefore replace its substrate and inhibit the 11β hydroxysteroid dehydrogenase enzyme causing an accumulation of glucocorticoids whose inactivation is delayed, with prolongation of the anti-inflammatory action;
- 2) inhibits the activation of the classical pathway of complement through a mechanism that depends on the molecular conformation of glycyrrhetic acid.

FUMARIC ACID 0,5%

Recently also admitted as active pharmaceutical ingredients for oral use, the compounds derived from fumaric acid have a certain effectiveness in combating the inflammatory process that it develops in psoriatic skin, blocking the production of proinflammatory cytokines.

ANTIFUNGAL ACTION

CICLOPIROX OLAMINE 0.5%

CPO has a strong antifungal and antibacterial action, it blocks the production of ATP (adenosine triphosphate), an energy reserve necessary for the vital functions of the fungal cell. Blocks carriers carrying leucine, valine and phenylamine, disrupting mRNA and protein synthesis. Furthermore, CPO binds and subtracts Fe^{3+} by inactivating cellular enzymes, causing the death of fungi and bacteria. Furthermore, CPO is recognized as having an important anti-inflammatory action, as it inhibits the inflammatory mediators originating from the arachidonic acid, by the action of 5α lipoxygenase.

TIOCONAZOLE 0,25%

Tioconazole is an active ingredient belonging to the category of imidazole derivatives with a broad spectrum of action which guarantees its therapeutic activity both against yeasts, dermatophytes and fungi and against Gram positive bacteria.

Like other antifungals, this active ingredient, applied topically, easily permeates the membranes of microorganisms, accumulating in the cytoplasm and inhibiting the enzyme 14α methyl lanosterol

demethylase, thus determining:

- Reduced synthesis of ergosterol, a key element for the structure and functioning of the plasma membrane;
- Accumulation of intermediate metabolites which reduce the activity of the various membrane enzymes, thus determining a significant reduction in the exchange and energy capacity.

All this is achieved without the active principle being absorbed systemically, thus limiting the appearance of side effects and adverse reactions.

Crinipan® PMC green 0,5%

(PROPANEDIOL CAPRYLATE)

Crinipan® PMC green is the first microactivated anti-dandruff active ingredient of natural origin with activity equivalent to climbazole. The new molecule combines efficacy and an intelligent mode of action: activated on the scalp by Malassezia enzymes, its activity is activated directly by the microorganism itself which causes dandruff. Being an ester, Propanediol Caprylate undergoes hydrolysis by Malassezia enzymes and is split into Propanediol and Caprylic Acid. The latter exercises a down regulation action on Malassezia, reducing its presence.

Seascalp™ biomarine ingredient 1%

(AQUA, PSEUDOALTEROMONAS FERMENT EXTRACT, SODIUM SALICYLATE)

The biomarine ingredient Seascalp™ is a biotechnological ingredient that helps minimize the accumulation of sebaceous lipids and strengthen the skin barrier function. The ingredient helps to naturally provide a healthy scalp with less itchiness, flaking and oiliness, while aiding in the day-to-day management of dandruff-prone scalps. Contains Sodium Salicylate with an inhibiting action on the proliferation of bacteria and fungi.

ANTIPRURIGINOUS ACTION

POLIDOCANOL 2,5%

For this ingredient, also used pharmacologically, it has been hypothesized that the antipruritic effect is due to a mild local anesthetic action.

ACETYL HEXAPEPTIDE 49 (DELISENS®) 0,5%

The innovative peptide DELISENS (Acetyl Hexapeptide-49) improves the skin condition generated following the release from the PAR 2 receptor (Proteinase Activated Receptor 2) of pro-inflammatory mediators, also helping to restore the integrity of the damaged skin barrier. In sensitive skins it increases hydration and improves the smoothness and softness of skin thickening and keloids. The active ingredient has a marked soothing, antipruritic and repairman.

	Other active ingredients ARGININ 2% Scalp strengthener. ROSEMARY ESSENTIAL OIL 0,05% Antibacterial and antifungal of natural origin.
HOW TO USE	ACUTE FORMS: Use twice a week on the affected part. CHRONIC FORMS: Use once a week on the affected part. Warnings: do not apply to open or fresh wounds, near the eyes and mucous membranes, skin treated with antibiotics or other topical products.
INDICATIONS	The product is indicated for the treatment of skin thickening and for the psoriatic forms of the scalp.